Test Plan — Leish! Aesthetic Marketplace

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1. Objectives

Prove that discovery, booking, authentication, provider operations and AI degradation work for supported actors; prevent unauthorized access and regression of the booking funnel; and provide a repeatable release gate.

2. Test levels

Level	Tool	Scope
Static	TypeScript	Compile-time correctness
Unit	Vitest	Schemas, transforms, dates, pricing, fallba
API integration	Vitest + Supertest	Routes, auth, status codes, ownership mat
Component	Vitest + Testing Library when available	Modal steps, portals, navigation and visible
E2E	Playwright or headed browser smoke	Guest booking, provider confirmation, deep
Manual accessibility	Keyboard, reduced motion, axe/Lighthouse	Focus, contrast, responsive states
Firestore emulator	Firebase emulator	Adapter round trips, rules and migration

3. Required test cases

ID	Case	Expected
T-001	Search by listing name/category	Matching results; no-result recovery
T-002	Open listing from card and map	Correct detail view; map failure does not block cards
T-003	Submit valid guest booking	201, pending status, receipt
T-004	Submit malformed date/email/required field	400 or client validation; no write
T-005	Booking API unavailable	Visible retryable error; no silent failure
T-006	Login/register round trip	Sanitized user and valid JWT
T-007	Invalid/expired JWT	401 and session recovery
T-008	Client reads own appointments	Own records only
T-009	Client accesses another booking	401/403-safe response; no data leak
T-010	Provider confirms own booking	200 and allowed status transition
T-011	Provider mutates another salon	403 and unchanged data
T-012	Service/profile CRUD	Validated, owner-scoped persistence
T-013	Review eligibility	Completed booking required
T-014	Gemini missing key/timeout/malformed JSON	Deterministic fallback and `source: fallback`
T-015	Lookbook to booking	Moodboard survives all three steps
T-016	Deep link refresh	App shell loads and correct view opens
T-017	Mobile keyboard/reduced motion	No trapped focus, no unusable horizontal overflow
T-018	Redeploy/persistence	Data remains after restart/deploy for selected backend

4. Environments and fixtures

- Local API: `npm run dev` with isolated test data.
- Test backend: temporary JSON store or Firestore emulator; never production data.
- Seed users: client and provider demo identities from the repository, with test-only credentials.
- AI: run both keyless deterministic tests and a controlled integration test when credentials are available.
- Browser viewports: 360px, 768px and 1280px; timezone tests include Asia/Kuala_Lumpur.

5. Release gates

```
npm run typecheck
npm test
npm run build
```

Production additionally requires the launch-runbook smoke checks, authz 401/403 matrix, and evidence that the configured store survives redeploy. A failing or skipped check is not a pass.

6. Current coverage and gaps

The repository already contains Vitest coverage for routing, analytics, color regression, store behavior, concierge and server/data-store paths. The remaining high-value gap is comprehensive route-by-route authorization and full browser E2E coverage, especially Workspace, image/video studios, physical-device accessibility, and durable Firestore deployment verification.